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"creator" : "ari.neko@cvut.cz",
"timestamp" : "09-07-2026_17:36:30",
"variant" : 1,
"subjectName" : "operating-systems",
"categories" : ["Deadlocks & Synchronization", "Kernel Architecture", "File Systems"],
"questions" : [{
"question" : "Which of the following is NOT one of the Coffman conditions required for a deadlock?",
"answers" : ["Circular Wait", "Preemption", "Hold and Wait", "Mutual Exclusion"],
"correctIndex" : 1
}, {
"question" : "What is a binary semaphore commonly called when used strictly as a locking mechanism?",
"answers" : ["Monitor", "Condition Variable", "Critical Section", "Mutex"],
"correctIndex" : 3
}, {
"question" : "What is a critical section inside multi-threaded application software?",
"answers" : ["A background process handling system initialization routines", "A protected kernel execution memory register row", "A critical database constraint tracking primary indexing relationships", "A block of code that accesses shared resources and must not be concurrently run"],
"correctIndex" : 3
}, {
"question" : "What happens during a hardware interrupt?",
"answers" : ["The CPU suspends its execution stream to run an interrupt service routine", "The active process is terminated and marked as a zombie", "Virtual memory maps flush physical hard drive tracks", "The operating system completely reboots its kernel"],
"correctIndex" : 0
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"question" : "What privilege layer does the operating system kernel run in on modern CPUs?",
"answers" : ["Kernel mode (Ring 0)", "User mode (Ring 3)", "Safe mode", "Hypervisor mode"],
"correctIndex" : 0
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"question" : "What is the primary purpose of a system call?",
"answers" : ["To provide an interface for user programs to request services from the kernel", "To compile source code files into machine code", "To manage graphical window rendering pipelines", "To clear CPU cache configurations automatically"],
"correctIndex" : 0
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"question" : "What is a major advantage of a microkernel compared to a monolithic design?",
"answers" : ["Higher system stability and isolation of corrupted drivers", "Faster execution

performance speeds", "Lower context switching overhead costs", "Smaller total storage size footprint"],
"correctIndex" : 0
}, {
"question" : "What component maps file paths to matching inode numbers in a file system directory?",
"answers" : ["Master boot records (MBR)", "Directory entries (dentries)", "Virtual file layers (VFS)", "File allocation tables (FAT)"],
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"question" : "What structural pattern does a Hard Link follow compared to a Symbolic Link?",
"answers" : ["Hard links delete the original file target when broken; symbolic links do not", "A Hard Link points directly to the same inode; a Symbolic Link points to a path string", "Symbolic links can only reference binary files; hard links handle folders", "Hard links require administrative privileges to execute"],
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"question" : "What is a major advantage of a journaling file system like ext4?",
"answers" : ["It encrypts all user credentials automatically", "It speeds up graphics card rendering rates", "It compresses text files to half their initial disk sector allocations", "It records changes in a log to prevent corruption during sudden power losses"],
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